

Savitribai Phule Pune University  
T.Y.BCA NEP  
T.Y.B.C.A Practical Examination OCT/NOV  
Subject Code CA-315 PECP Sem-V  
Subject Name- Lab course on CA-314-PEC(Cyber Security)

Duration: 3 Hours

Maximum Marks: 35

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**Q1) A** Explain the CIA triad (Confidentiality, Integrity, Availability) [10 marks]  
with suitable examples.

**Q2) B** Write a Python program to simulate a phishing login page that [20 marks]  
collects username and password and stores them in a file.

OR

**Q2) B** Write a Python program to implement a dictionary attack using a [20 marks]  
list of common passwords to check whether a target password  
exists in the list.

**Q3)** Viva [05 marks]

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**Q1) A** Differentiate between virus, worm, and Trojan horse. [10 marks]

**Q2) B** Write a Python program to detect spam emails using predefined keywords. [20 marks]

OR

**Q2) B** Write a Python program to encrypt user input text using AES encryption [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** What is phishing? Explain the step-by-step algorithm to identify phishing emails. [10 marks]

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**Q2) B** Write a Python program to shift each alphabet by 2 positions. [20 marks]

OR

**Q2) B** Write a Python program to simulate a phishing login page for educational purposes.. [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Explain Brute Force Attacks with an example, detailing Simple, Dictionary, and Hybrid brute force types. [10 marks]

**Q2) B** Write a Python program to demonstrate IP tracking simulation by storing fake IP addresses and printing location mapping using a dictionary [20 marks]

OR

**Q2) B** Write a Python program to check password strength based on basic security rules. [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Write short notes on network security threats, including Man-In-The-Middle (MITM) and Packet Sniffing attacks. [10 marks]

**Q2) B** Write a Python program to demonstrate IP tracking (simulation):Store fake IP addresses and Print location mapping (use dictionary) [20 marks]

OR

**Q2) B** Write a simple Python program to hide a text message inside an image using steganography and display the hidden message again [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Explain the key behaviors of cyber stalkers and discuss the misuse of social networking features [10 marks]

**Q2) B** Write a Python program to check password strength: Length  $\geq 8$  and password must contain digits and special characters [20 marks]

OR

**Q2) B** Write a Python program to implement Caesar Cipher for both encryption and decryption of a given text. [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Define Denial of Service (DoS) attack and explain its working principles along with SYN flood and HTTP flood attack types [10 marks]

**Q2) B** Write a Python program using Scapy to detect a possible DoS attack by counting packets received from each IP address and generating an alert when traffic exceeds a threshold [20 marks]

OR

**Q2) B** Write a Python program to implement Caesar Cipher for both encryption and decryption of a given text [20 marks]

**Q3)** Viva [05 marks]

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- Q1) A** Define Steganography and distinguish it clearly from Cryptography [10 marks]
- Q2) B** Write a Python program that accepts a sentence from the user and encrypts it using Caesar Cipher [20 marks]

OR

- Q2) B** Write a Python program to implement a dictionary attack using a list of common passwords to check whether a target password exists in the list [20 marks]
- Q3)** Viva [05 marks]

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**Q1) A** Explain the key generation, encryption, and decryption steps of the RSA algorithm with mathematical formulas [10 marks]

**Q2) B** Write a Python program using PyCryptodome to encrypt and decrypt the message "HELLO WORLD" using AES with a predefined 16-byte key [20 marks]

OR

**Q2) B** Write a Python program to encrypt user input text using AES encryption [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Compare AES with RSA based on type, key size, speed, and security, and explain key AES applications. [10 marks]

**Q2) B** Write a Python program to perform secure message communication using AES encryption and decryption [20 marks]

OR

**Q2) B** Write a Python program to generate MD5, SHA-1, and SHA-256 hashes of a file and verify its integrity. [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Explain the "Cover Object" and "Stego-Object" in the context of cyber security. [10 marks]

**Q2) B** Create a forensic image of a USB drive using FTK Imager, save it in E01 format, and generate a hash report [20 marks]

OR

**Q2) B** Write a Python program using Scapy to detect a possible DoS attack by counting packets from each IP address and generating an alert when traffic exceeds a limit [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** What specific digital evidence can be obtained from browser history analysis in digital forensics? [10 marks]

**Q2) B** Write a simple Python program to hide a text message inside an image and display the hidden message again. [20 marks]

OR

**Q2) B** Calculate and encrypt the number  $M = 5$  using the RSA algorithm given  $p=3$ ,  $q=11$ , and  $e=7$ . [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Write a program to encrypt a message using Caesar Cipher. [10 marks]

**Q2) B** Implement a Python program to detect a possible DoS/DDoS attack by monitoring network traffic [20 marks]

OR

**Q2) B** Write a Python program to implement AES encryption and decryption using a custom secret key. [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Explain Man-In-The-Middle (MITM) attacks and Packet Sniffing along with their preventive measures. [10 marks]

**Q2) B** Write a Python program to simulate a SHA-256 hash generator that accepts user input text and computes its corresponding 256-bit hash value. [20 marks]

OR

**Q2) B** Write a program to display ASCII values of characters before encryption [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Explain the concepts of Salami Attack, Data Diddling, Web Jacking, and Email Bombing as forms of cyber offenses [10 marks]

**Q2) B** Write a Python program to simulate a Brute Force Attack that generates all 3-character combinations from a given set of characters to discover a target password. [20 marks]

OR

**Q2) B** Write a Python program to demonstrate IP tracking (simulation):Store fake IP addresses and Print location mapping (use dictionary) [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Describe the different types of Distributed Denial of Service (DDoS) flooding attacks. [10 marks]

**Q2) B** Write a program using both ord() and chr() functions for Caesar Cipher [20 marks]

OR

**Q2) B** Write a program to implement RSA algorithm.(Generate public and private keys, encrypt a message, and decrypt it.) [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Write short notes on network security threats, including Man-In-The-Middle (MITM) and Packet Sniffing attacks [10 marks]

**Q2) B** Write a Python program to simulate packet sniffing and alert for sensitive keywords like password and PIN [20 marks]

OR

**Q2) B** Write a Python program to implement Caesar Cipher for both encryption and decryption of a given text. [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Define Steganography and distinguish it clearly from Cryptography. [10 marks]

**Q2) B** Write a Python program that accepts a sentence from the user and encrypts it using Caesar Cipher [20 marks]

OR

**Q2) B** Write a Python program to implement AES encryption in ECB mode using the pycryptodome library to pad and encrypt a 16-byte message. [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** What is Digital Evidence? Explain the role and purpose of Write Blockers, FTK Imager, and EnCase Imager in forensic acquisition.. [10 marks]

**Q2) B** Write a Python program that accepts a sentence from the user and encrypts it using Caesar Cipher [20 marks]

OR

**Q2) B** Write a program to implement RSA algorithm.(Generate public and private keys, encrypt a message, and decrypt it.) [20 marks]  
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**Q1) A** Describe the different types of Distributed Denial of Service (DDoS) flooding attacks.. [10 marks]

**Q2) B** Write a Python program to detect spam emails using predefined keywords [20 marks]

OR

**Q2) B** Write a program to display ASCII values of characters before encryption. [20 marks]

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**Q1) A** What is forensic image integrity verification? Explain the use of MD5 and SHA-1 hashes [10 marks]

**Q2) B** Explain browser history analysis from a forensic image, including visited websites and downloaded files [20 marks]

OR

**Q2) B** Write a program to encrypt user input text using AES.(Take plaintext from the user and encrypt it.). [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** List any three types of files that can be used as carriers in steganography [10 marks]

**Q2) B** Write a program to verify file integrity using MD5, SHA-1, and SHA-256 hash values. [20 marks]

OR

**Q2) B** Write a Python program to check password strength: Length  $\geq 8$  and password must contain digits and special characters [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Discuss the practical applications and use cases of the RSA algorithm, including Digital Signatures [10 marks]

**Q2) B** Write a Python program using Scapy to detect a possible DoS attack by counting packets from each IP address and generating an alert when traffic exceeds a limit [20 marks]

OR

**Q2) B** Write a Python program to simulate a phishing login page that collects username and password and stores them in a file [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** What is a Man-In-The-Middle (MITM) attack? Explain common techniques used. [10 marks]

**Q2) B** Write a Python program to demonstrate IP tracking (simulation):Store fake IP addresses and Print location mapping (use dictionary) [20 marks]

OR

**Q2) B** Write a simple Python program to hide a text message inside an image using steganography and display the hidden message again . [20 marks]

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**Q1) A** What is phishing? Explain the step-by-step algorithm to identify phishing emails. [10 marks]

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**Q2) B** Write a simple Python program to hide a text message inside an image and display the hidden message again. [20 marks]

OR

**Q2) B** Write a Python program to simulate a phishing login page for educational purposes.. [20 marks]

**Q3)** Viva [05 marks]

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**Q1) A** Write a Python program to decrypt a Caesar Cipher message.. [10 marks]

**Q2) B** Write a program to implement RSA algorithm.(Generate public and private keys, encrypt a message, and decrypt it.) [20 marks]

OR

**Q2) B** Perform a digital forensic investigation by creating a forensic image, verifying integrity, analyzing files/deleted data, and preparing a report. [20 marks]

**Q3)** Viva [05 marks]